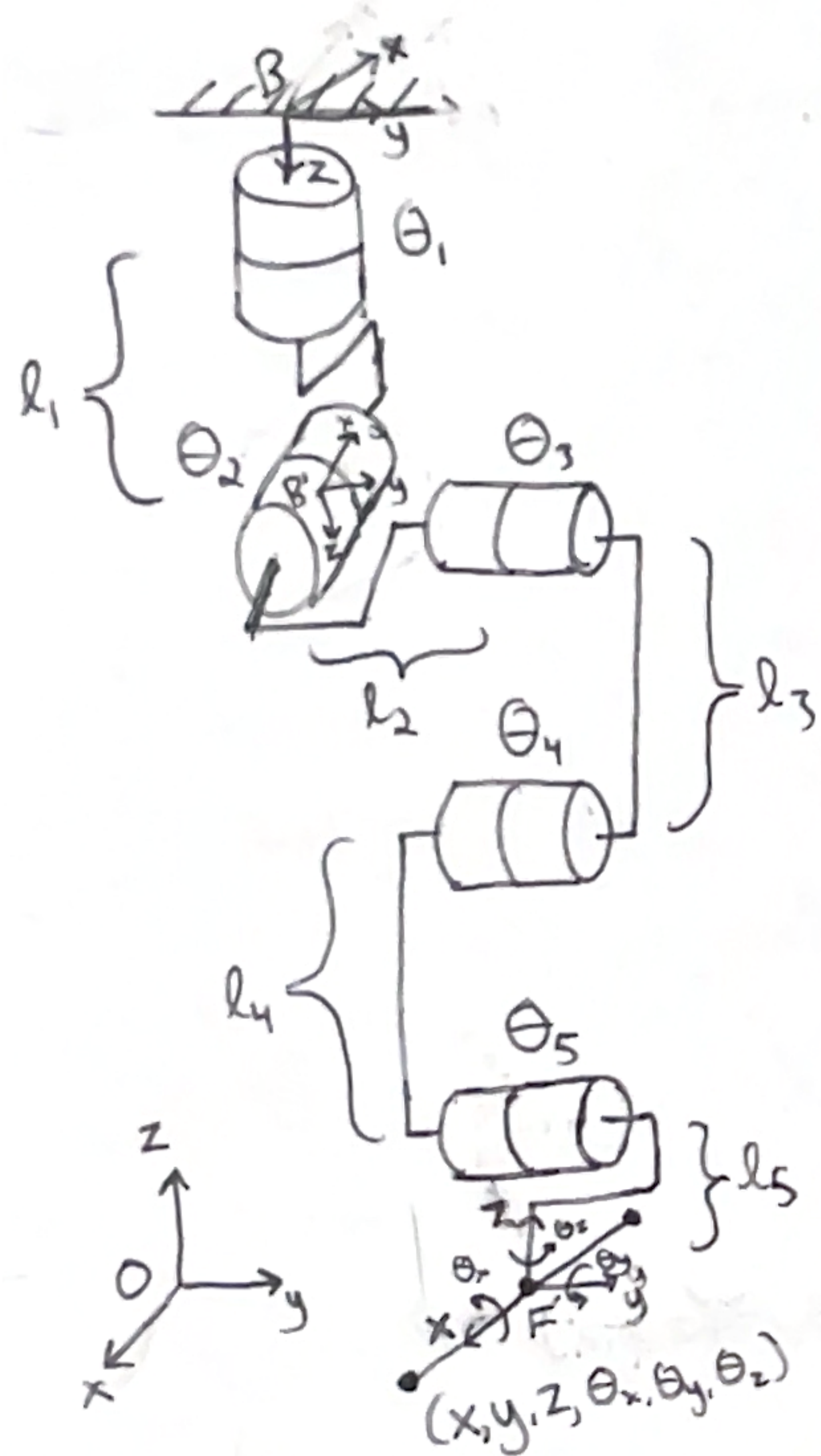




x - changed by $\theta_1, \theta_3, \theta_4, \theta_5$

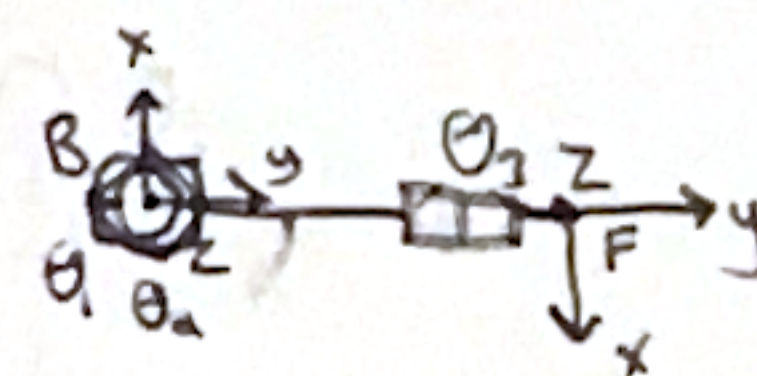
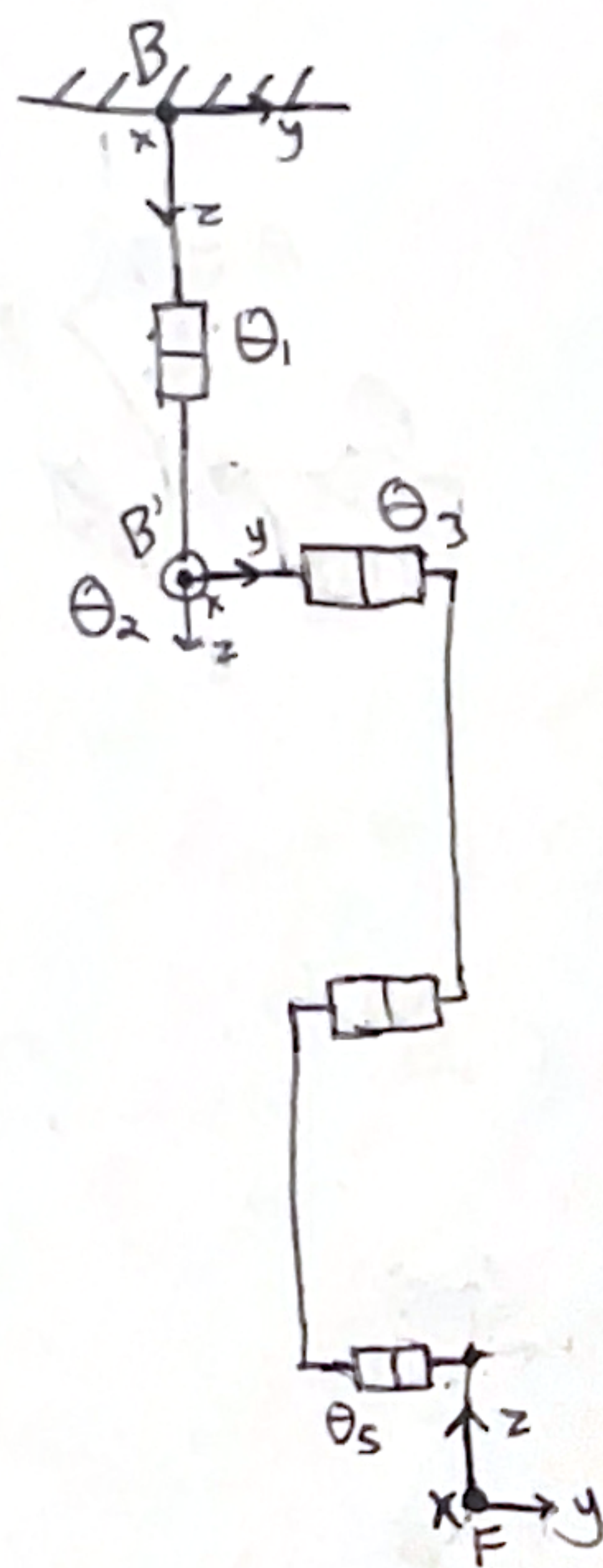
y - changed by θ_1, θ_2

z - changed by $\theta_2, \theta_3, \theta_4, \theta_5$

θ_x - changed by θ_2

θ_y - changed by $\theta_3, \theta_4, \theta_5$

θ_z - changed by θ_1



$$F_x = -B_x = -B'_x$$

$$F_y = B_y + l_2 = B'_y + l_2$$

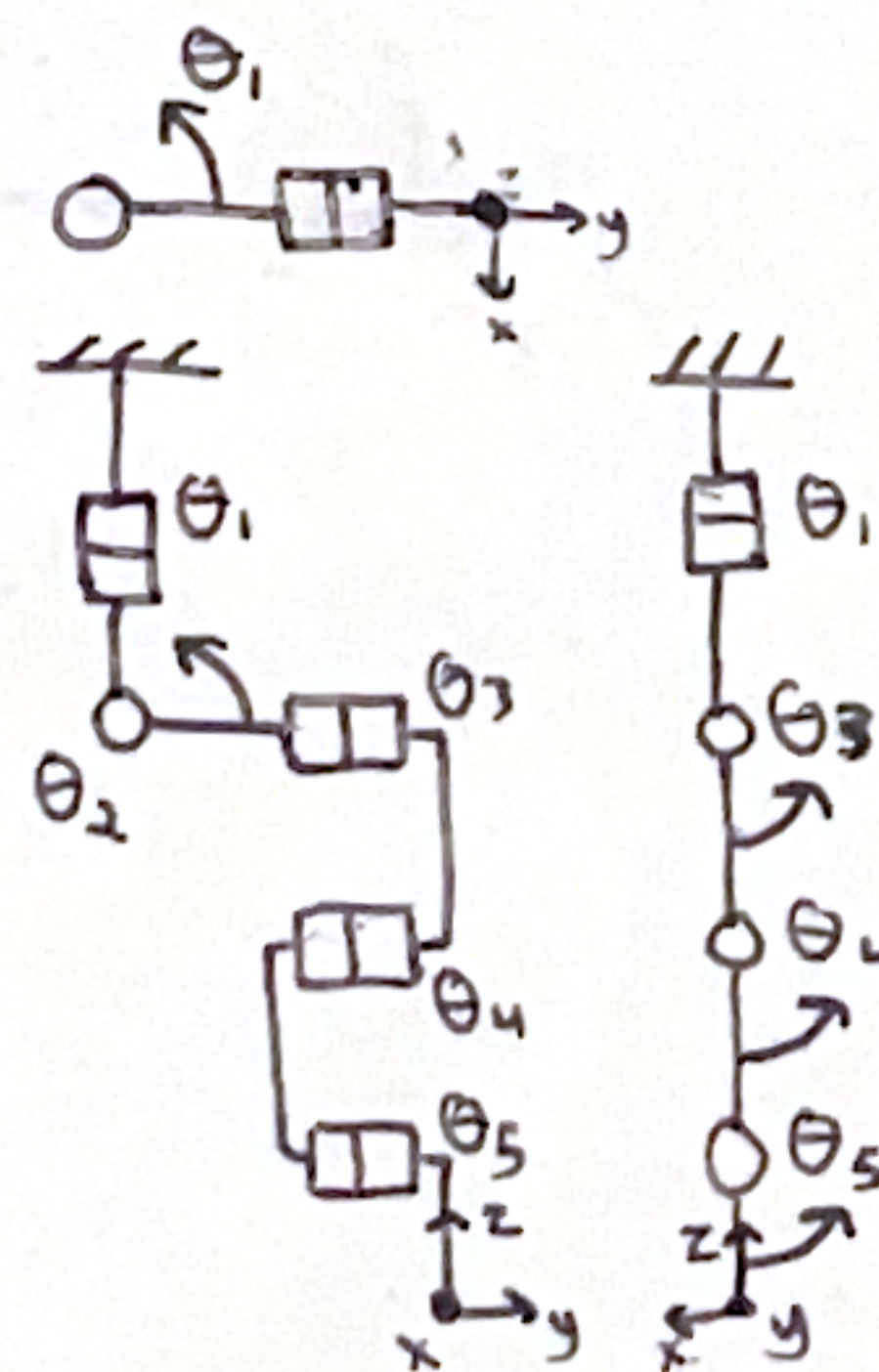
$$F_z = -B_z + l_1 + l_3 + l_4 + l_5 \\ = -B'_z + l_3 + l_4 + l_5$$

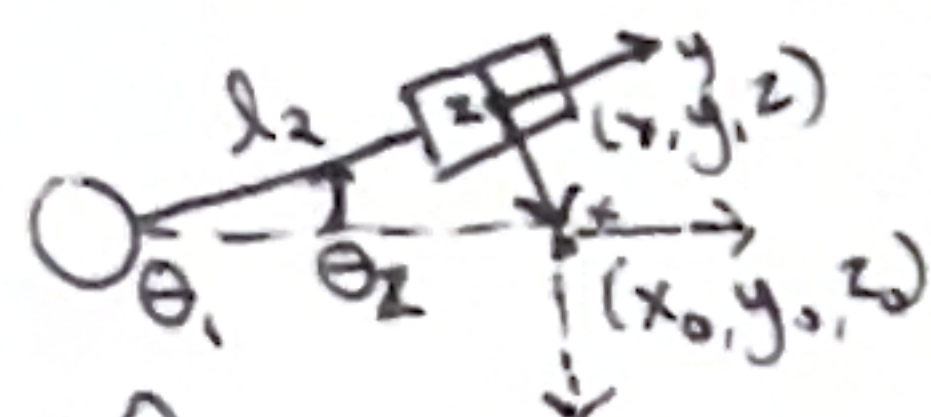
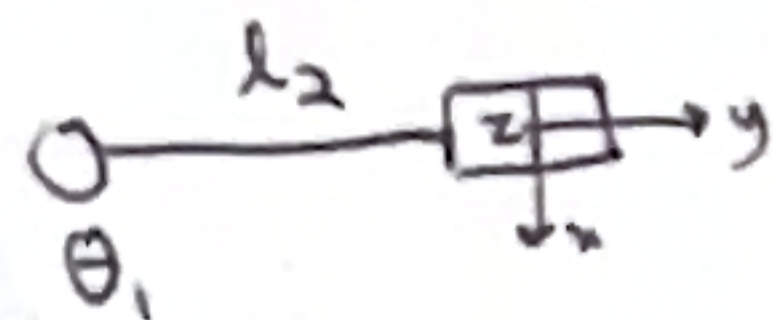
if $\theta_y = 0$

$$F'_x = F_x$$

$$F'_y = F_y - l_5 \sin \theta_2$$

$$F'_z = F_z - l_5 \cos \theta_2$$

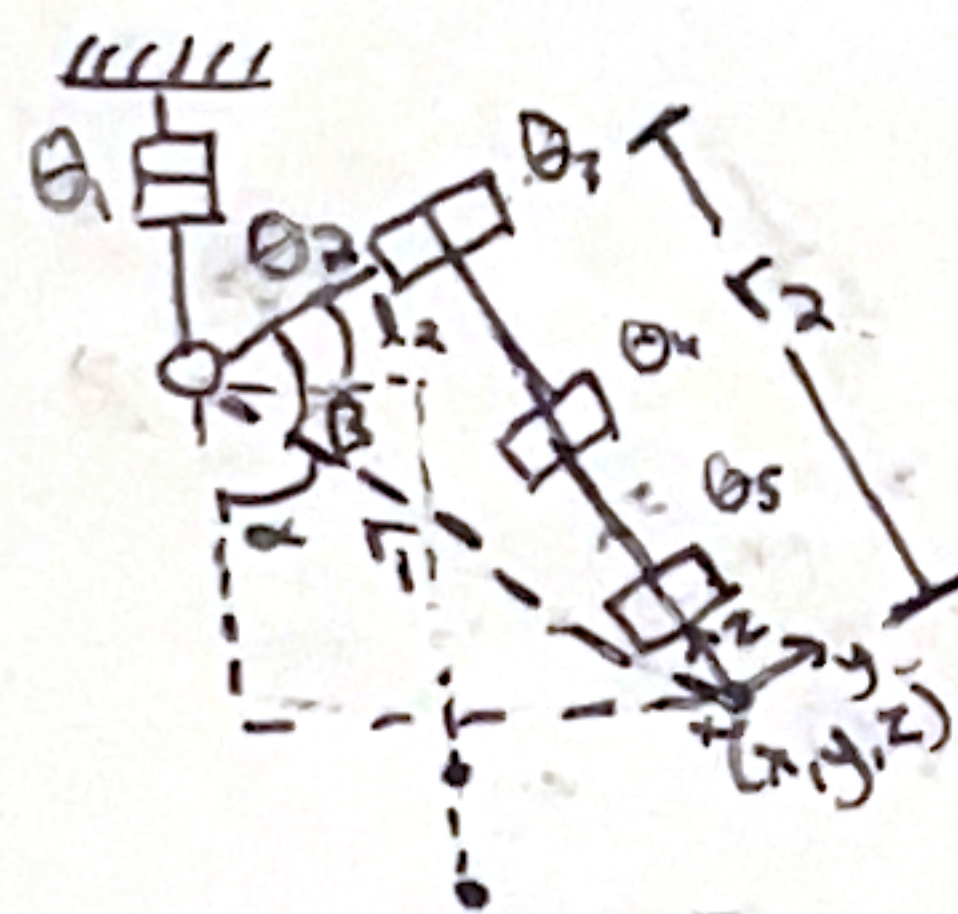
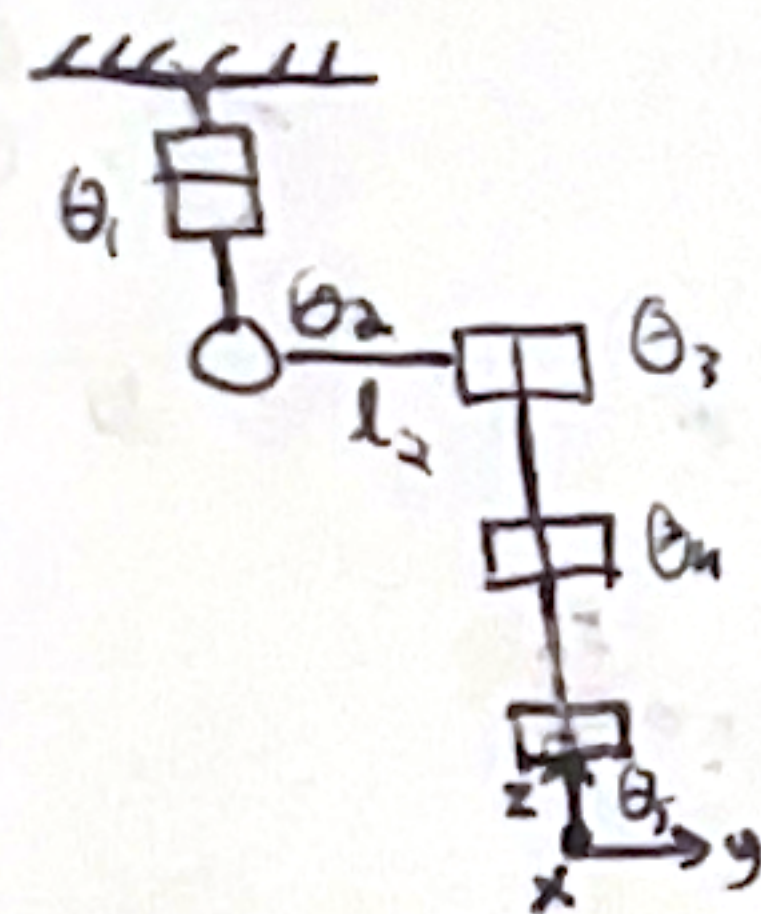




$$\theta_1 = \theta_2$$

$$x = -(x_0 \cos(-\theta_2) + y_0 \sin(-\theta_2))$$

$$y = -x_0 \sin(-\theta_2) + y_0 \cos(-\theta_2)$$

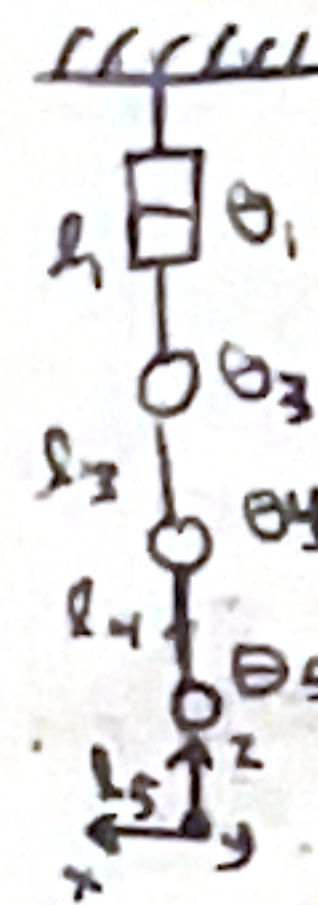


$$r_1 = \sqrt{y^2 + z^2}$$

$$\alpha = \cos^{-1}\left(\frac{z}{r_1}\right)$$

$$\beta = \cos^{-1}\left(\frac{l_2}{r_1}\right)$$

$$\theta_2 = \alpha + \beta - \frac{\pi}{2}$$



$$r_2 = r_1 \sin \beta$$

$$r_3 = \sqrt{x^2 + (r_2 - l_3)^2}$$

$$r_3^2 = l_3^2 + l_4^2 - 2l_3l_4 \cos \delta \rightarrow \delta = \cos^{-1}\left(\frac{l_3^2 - l_3^2 - l_4^2}{-2l_3l_4}\right)$$

$$\theta_3 = \tan^{-1}\left(\frac{x}{r_2 - l_3}\right)$$

$$\frac{\sin \gamma}{l_4} = \frac{\sin \delta}{r_3} \rightarrow \gamma = \sin^{-1}\left(\frac{l_4 \sin \delta}{r_3}\right)$$

$$\theta_3 = \theta_3 + \gamma$$

$$\theta_4 = -(\pi - \delta)$$

$$\theta_5 = -\theta_3 - \theta_4 + \theta_y$$